

NuScale Power Corporation

First NRC-Licensed SMR Design — 77 MWe VOYGR Light Water Modular Reactor

PhD Due Diligence Report | BLRI-DD-SMR-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

NuScale VOYGR: The Only NRC-Approved Small Modular Reactor in the United States

NuScale Power holds the first and only NRC-approved small modular reactor design certification (DC/COL-003) in the United States — a historic milestone granted in January 2023. The VOYGR-12 configuration is 12 modules × 77 MWe = 924 MWe total, scalable down to VOYGR-4 (308 MWe) or VOYGR-6 (462 MWe). The passive safety system requires no active cooling during accidents — a fundamental improvement over Generation II/III reactors. In Seldon ξ -variable terms: NuScale's scalable modular approach could democratise nuclear power for mid-sized utilities, developing nations, and industrial facilities that cannot support a gigawatt-scale plant.

NuScale's challenge is existential: the UAMPS Carbon Free Power Project (CFPP) cancellation in 2023 revealed a fatal economics problem — as project costs escalated from \$58/MWh to \$89/MWh, customers withdrew. NuScale must find a viable cost reduction pathway or a customer base (military, international, industrial) where cost is secondary to energy security.

Source: moomoo OpenD [2026-08-13] · NRC Design Certification DC/COL-003 · UAMPS CFPP cancellation announcement

PART II — QUALITATIVE ANALYSIS

Post-UAMPS Strategy: International + Military Markets

After the CFPP cancellation, NuScale pivoted to international markets: Romania (RATEN MOE agreement), Poland (PKN Orlen), South Korea (KEPCO collaboration), and South Africa. These partnerships are non-binding MoUs. The US Department of Defense has evaluated VOYGR for military base power. NuScale's international pivot is logical — energy security in Europe and Asia after Russia's Ukraine invasion makes nuclear economically viable even at higher cost per MWh.

The Cost Problem: Factory Manufacturing Not Yet Proven

NuScale's SMR cost reduction thesis requires factory-manufactured modules shipped by rail to sites. This modular manufacturing has never been demonstrated at scale. The unit cost trajectory depends on learning curve effects across 10+ units — a chicken-and-egg: first customers pay a premium that makes the economics non-competitive, depressing demand for subsequent units.

Source: NuScale Power 10-K EDGAR · DOE SMR programme · IEA Nuclear report

PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]

Price: \$9.890 | Prev Close: \$9.590 | Mkt Cap: \$4.059B
 EPS TTM: -\$2.170 | Net Income: -\$890.5M | Div Yield: 0%
 52w: \$7.210–\$57.420 | PE TTM: -4.50x | Shares: 410.4M

Metric	Value	Source
Price (2026-08-13)	\$9.890	moomoo OpenD
Market Cap	\$4.059B	moomoo OpenD
EPS TTM	-\$2.170	moomoo OpenD
Net Income TTM	-\$890.5M	moomoo OpenD
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PART IV — COMPETITIVE LANDSCAPE

Competitor	Type	Mkt Cap	vs SMR
Oklo (OKLO)	Fast fission microreactor	\$8.577B	OKLO smaller scale; SMR NRC-approved only LWR SMR
TerraPower (private)	Sodium-cooled SMR (Natrium)	Private (>\$1B)	Wyoming demo 2030; SMR has NRC head start
Rolls-Royce SMR (private)	UK LWR SMR (470 MWe)	Private	UK focus; SMR US-licensed advantage
BWRX-300 (GE-Hitachi)	Boiling water SMR	N/A (subsidiary)	BWRX licensing in progress; SMR already approved

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The Romanian Energy Ministry Official (COMPOSITE)

A Romanian energy policy director evaluating NuScale's VOYGR for a 462 MWe plant at Doicești. 'After Chernobyl, Romania never completed its second Cernavoda unit. SMR gives us a politically acceptable nuclear pathway — smaller, modular, passive safety. EU taxonomy classification of nuclear as green energy makes financing possible.' Represents the European energy security market where SMR's economics (\$89/MWh) are competitive vs. gas at \$120/MWh.

COMPOSITE / ILLUSTRATIVE — drawn from European nuclear policy patterns

FINDING F-5-B: Portrait 2: The DOD Energy Security Analyst (COMPOSITE)

A Pentagon energy analyst evaluates VOYGR-4 (308 MWe) for a major Pacific military installation. 'Our bases in Guam and Diego Garcia need resilient power independent of civilian grid. A 308 MWe SMR provides 10 years of guaranteed power at predictable cost — the energy security calculus overrides the cost premium.' Represents the military market where SMR's cost disadvantage is irrelevant vs. energy security value.

COMPOSITE / ILLUSTRATIVE — drawn from US military energy security patterns

FINDING F-5-C: Portrait 3: The Utility CFO Who Cancelled Their UAMPS Share (COMPOSITE)

A Utah municipal utility CFO explains the UAMPS withdrawal: 'When the projected electricity price rose from \$58 to \$89/MWh, our board couldn't justify it against gas and solar at \$45/MWh. NuScale has a real product — but they need to demonstrate cost containment before we re-engage.' Represents the fundamental commercial economics problem NuScale must solve to access the US utility market.

COMPOSITE / ILLUSTRATIVE — drawn from UAMPS withdrawal documentation patterns

PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: HH1: SMR economics remain uncompetitive — survival risk

EVIDENCE FOR:

UAMPS cancellation revealed all-in cost of \$89/MWh; gas + solar at \$40-50/MWh; no US utility willing to be first mover at premium; cash burn of \$890M/year is unsustainable pre-revenue

EVIDENCE AGAINST:

International markets (Romania, Poland, South Korea) have higher gas prices making SMR competitive; DOE loan guarantees reduce financing cost; military market has no price ceiling

QUANTITATIVE TEST

P(NuScale running out of cash without new orders before 2027)=40%

VERDICT: BEAR — commercial economics are NuScale's Achilles heel

Implication: Monitor quarterly cash position; international MoU conversion to binding contracts

HH2: Romania VOYGR project signs binding EPC contract

EVIDENCE FOR:

Romania's RATEN MOE has signed a preliminary agreement; EU taxonomy green nuclear enables EIB financing; Romanian government has political motivation for energy independence

EVIDENCE AGAINST:

MoU is non-binding; Romanian government budget constraints; construction timeline 10+ years even with contract

QUANTITATIVE TEST

P(Romania binding EPC contract within 18 months)=20%

VERDICT: BULL — first binding international contract would re-rate SMR significantly

Implication: Monitor Romanian Ministry of Energy announcements; EIB nuclear financing applications

HH3: DOD emergency nuclear programme selects NuScale

EVIDENCE FOR:

Post-Ukraine energy crisis; Taiwan Strait tensions accelerating US military energy independence; Pacific island bases need resilient power; NuScale is the only NRC-licensed US option

EVIDENCE AGAINST:

DOD procurement cycles are 5-7 years; Congressional appropriation required; NuScale may not survive long enough to win contract

QUANTITATIVE TEST

P(DOD SMR contract >\$500M for NuScale by 2027)=15%

VERDICT: BULL — DOD contract would validate technology and provide non-dilutive revenue

Implication: Monitor Pentagon Advanced Nuclear Energy for Bases (ANEB) programme budget

HH4: Cost reduction roadmap fails — SMR never reaches commercial viability

EVIDENCE FOR:

Factory manufacturing learning curve requires 10+ unit orders; first unit always expensive; no utility willing to pay premium; SMR enters negative demand spiral

EVIDENCE AGAINST:

DOE cost-share programme funds first N units at below-market rates to establish learning curve; international demand provides alternative first-mover customers

QUANTITATIVE TEST

P(SMR never achieving \$60/MWh economics before 2035)=35%

VERDICT: BEAR — cost trajectory is existential

Implication: Monitor DOE SMR programme funding; NuScale engineering cost reduction announcements

HH5: HH5: Acquisition by Korean KEPCO or French EDF

EVIDENCE FOR:

KEPCO has collaborated with NuScale; Korean government seeking US nuclear partnership; EDF wants US SMR market access via NuScale licence; acquisition at 30-50% premium to market

EVIDENCE AGAINST:

CFIUS review would scrutinise Korean state acquisition of NRC-licensed US nuclear technology; DOE national security concerns

QUANTITATIVE TEST

$P(\text{NuScale acquired by Asian/European utility before 2027}) = 20\%$

VERDICT: BULL — acquisition would rescue shareholders and accelerate international deployment

Implication: Monitor KEPCO/EDF partnership announcements; NuScale board M&A discussions

PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: Romania signs binding EPC contract (P = 18%)

RATEN MOE converts MoU to binding construction contract; first VOYGR-6 at Doicești confirmed; EU financing secured; SMR re-rates to \$25+

Driver: Romanian Ministry of Energy announcements; EU taxonomy nuclear financing

Falsifier: Romanian budget crisis; contract delayed past 2027

SCENARIO 2: DOD contract for Pacific base SMR (P = 12%)

Pentagon awards NuScale contract for Guam/Hawaii installation; national security budget bypasses cost sensitivity; \$500M+ contract

Driver: Congressional defence appropriations; Pentagon ANEB programme

Falsifier: DOD selects solar+battery+diesel instead; SMR cost too high

SCENARIO 3: Base case: survival mode on international MoU pipeline (P = 30%)

Romania/Poland MoUs maintained; South Korea collaboration continues; no binding contracts; equity raise to fund 2 more years; stock \$6-12

Driver: Quarterly cash burn; MoU status updates

Falsifier: Cash runs out; forced restructuring or dilutive raise below \$5

SCENARIO 4: South Korea KEPCO deployment agreement (P = 15%%)

KEPCO selects NuScale for Korean domestic SMR programme; licensing partnership; \$1B+ commercial agreement; stock re-rates

Driver: KEPCO SMR strategy announcements; NRC/KINS bilateral cooperation

Falsifier: Korea develops proprietary SMR (SMART, BWRX) without NuScale

SCENARIO 5: Equity raise at \$6 triggers selloff to \$7.21 low (P = 35%%)

NuScale raises equity at discount to fund operations; dilution concern; stock retests 52-week low

Driver: Quarterly cash position; equity raise announcement

Falsifier: Binding contract announced before raise; raise cancelled

SCENARIO 6: DOE loan guarantee for first US demonstration unit (P = 25%%)

DOE LPO approves \$1-2B loan guarantee for first US NuScale unit; cost of capital drops; commercial economics improve

Driver: DOE LPO advanced nuclear programme; NuScale application status

Falsifier: LPO selects competing advanced nuclear technology

SCENARIO 7: NuScale academic/research market entry (P = 20%%)

Major universities (MIT, Stanford, Georgia Tech) deploy VOYGR for campus power + research reactor replacement; non-commercial market entry provides reference site

Driver: University energy master plan announcements; NRC academic reactor licensing

Falsifier: Universities prefer solar+storage; nuclear perception risk on campus

SCENARIO 8: Collaboration with Saudi Arabia (NEOM) (P = 10%%)

Saudi Arabia selects NuScale for NEOM mega-city power; non-OECD customer with capital to absorb premium pricing; \$5B+ programme

Driver: NEOM energy procurement; Saudi nuclear programme

Falsifier: Saudi Arabia selects Chinese/Korean competitor technology

SCENARIO 9: NuScale merges with Oklo or TerraPower (P = 8%%)

Regulatory merger of two pre-revenue advanced nuclear companies; combined entity with multiple technologies; DOE encourages consolidation

Driver: Nuclear industry M&A; DOE advanced reactor programme signals

Falsifier: DOE anti-consolidation; combined entity still struggles with economics

SCENARIO 10: Bull case: First commercial VOYGR operational by 2030 (P = 5%%)

Romania unit construction starts 2026; first power by 2030; commercial proof-of-concept; multiple follow-on orders; stock \$40+

Driver: All milestone fires; binding contract + financing + construction start

Falsifier: Timeline extends to 2032+; cost overruns; SMR thesis unfulfilled

PART VIII — SELDON VARIABLE DEEP DIVE

Variable	SMR Impact	Direction	Magnitude
K	SMR enables nuclear for mid-tier nations/utilities	↑ Long-term	K+ if commercialised
ξ	Scalable nuclear → energy equity for nations without large nuclear programmes	↑ Long-term	ξ+ decade horizon
Ω	Domestic US nuclear supply reduces energy geopolitical dependency	↓ Reduces	Ω- moderate (licensed only)
I	NRC certification validates US nuclear regulatory leadership	↑ Mild	I+ moderate

PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
Commercial economics non-competitive	HIGH (40%)	CRITICAL	International markets + military pricing tolerance
Cash burn — pre-revenue	HIGH (35%)	CRITICAL	DOE loan guarantee +

insolvency			international contract pursuit
No binding contracts by 2027	HIGH (40%)	HIGH	Romania/Korea MoU conversion is key milestone
UAMPS precedent chills US utility demand	HIGH (confirmed)	HIGH	International pivot is correct strategic response
NRC design certification expiry or revision	LOW (10%)	HIGH	Design certification valid; incremental amendments only

CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: SPECULATIVE — NRC ADVANTAGE IS REAL; COMMERCIAL ECONOMICS ARE THE CRISIS

NuScale at \$9.890 (mkt cap \$4.059B) holds the only NRC-approved SMR design certificate in the US — a structural advantage no competitor can replicate for 5+ years. The commercial economics problem (UAMPS cancellation at \$89/MWh) is real and severe. Survival hinges on Romania binding contract OR DOD programme OR DOE loan guarantee. Without one of these by 2027, cash burn risk is acute. Seldon ξ+ long-duration (modular nuclear for developing nations). HUMAN_ONLY_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,555ch.